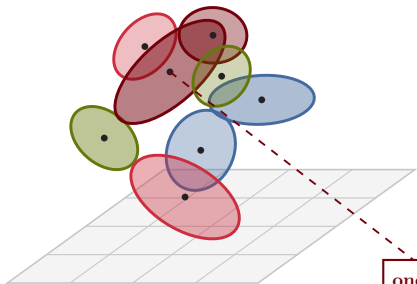
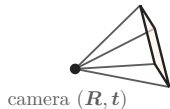


3-D Gaussian Splatting

explicit radiance field: rasterize & α -blend a cloud of 3-D Gaussians

Anisotropic 3-D Gaussians

$$\text{scene} = \{(\mu_g, \Sigma_g, \alpha_g, c_g)\}_{g=1}^N$$



one splat g
 $\Sigma_g = RSS^T R^T$
opacity α_g , color c_g (SH)
ellipsoid = $\sqrt{\lambda_i}$ semi-axes

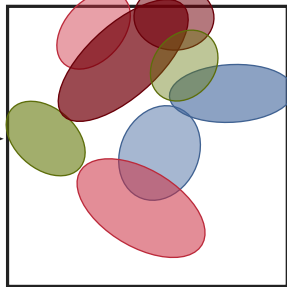
project to screen

$$\Sigma' = JW\Sigma W^T J^T$$

differentiable
tile rasterizer
sort by depth
 α -blend (over)
 $C = \sum_i c_i \alpha_i \prod_{j<i} (1 - \alpha_j)$

Rendered view

α -blended image $C(u)$



per-pixel over-composite of overlapping splats

$$\mathcal{L} = \|C - C^{\text{gt}}\|_1 + \dots$$

$\nabla_{\mu, \Sigma, \alpha, c} \mathcal{L}$ (back to every splat)